

SOURAV PATTANAIK

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MS graduate in Biology with a multidisciplinary research experience spanning systems biology, structural bioinformatics, molecular simulations, molecular biology, and chemistry. I am particularly interested in integrative biology workflows that combine computation with experiments to study biological phenomena and enable rational therapeutic discovery.

EDUCATION

2021 – 2026	Integrated Master of Science in Biology (Minor degree in Chemistry) National Institute of Science Education and Research (NISER), India Integrated CGPA: 8.40 / 10.0 Average Bachelor's (First Six semesters) SGPA: 8.02 / 10.0 Average Master's (Last four semesters) SGPA: 8.96 / 10.0
2021	12th Boards (CBSE) DAV School — 94.6%
2019	10th Boards (ICSE) Chinmaya Vidyalaya (E.M.) — 93.2%

PUBLICATIONS

Peer-reviewed Publications

Rana S., **Pattanaik S.**, Thakur R., Joshi V., Jindal N., Tiwari R. R., & Sahoo G. C. (2026). Computational design and validation of a multi-epitope vaccine candidate against structural proteins of Langya virus. *Letters in Drug Design & Discovery*, 100296. <https://doi.org/10.1016/j.lidd.2026.100296> **Accepted**

Conference Abstracts

Pattanaik, S., Konkimalla, V. B. Directional Biological Network (DBN) Analysis Pro: Automated DBN construction, target prioritization, and pathway enrichment analysis. Abstract submitted to *IndiBioCom 2026*. Registration ID: IBC-2026-16316D33. **Submitted**

RESEARCH EXPERIENCE

2024 – 2026	MASTER'S THESIS Supervisor: Dr. V Badireenath Konkimalla, NISER, India <i>"Integrating network-based drug discovery and stimulus-responsive drug carrier for treating testicular cancer through an in silico approach."</i> Received an average of 9.25 / 10 grade - Developed DBN Analysis Pro, a web tool to integrate transcriptomic, pathway, and protein-protein interaction data for mechanistic prioritization of therapeutic targets. - Designed a novel dual-stimulus-responsive drug carrier using Intrinsically Disordered Proteins (IDPs) and studied its conformational dynamics using CG simulations. - Leveraged atomistic simulations for studying protein-ligand interactions after virtual screening for a potential drug molecule among 1.8 million molecules in the PubChem Database. - Methods: AlphaFold, GROMACS, CG (MARTINI)/Atomistic (CHARMM) MD Simulation, Pharmacophore modeling, Protein-DNA Docking, Network analysis.
May – July 2025	SUMMER FELLOWSHIP Dr. Anbazhagan Kolandaswamy, RRMCH Bengaluru <i>"Discovery of Biomarkers for Sepsis Disease through comparative analysis of gene expression profiles."</i> - Optimized RNA extraction protocols and processed 52 patient blood samples, while maintaining high-integrity RNA extraction for quantitative profiling of 15 genetic targets (mRNA/miRNA). - Identified 6 statistically significant potential biomarkers (P-values < 0.05). - Methods: qPCR, RT-PCR, Biostatistical analysis, Flow cytometry for immunophenotyping, FlowJo software for analysis, gel electrophoresis.
Jan – Mar 2025	REMOTE COLLABORATION Dr. Karoline Faust, KU Leuven <i>"Beta-testing of a new database associated with Systems Microbiology."</i> - Extracted, structured, and interpreted growth data of 10 different bacterial species. - Contributed to Beta-testing of the new database. - Acknowledged in research pre-print (doi:10.1101/2025.10.13.682118).
May – July 2024	SUMMER INTERNSHIP Dr. Sindhuprava Rana, ICMR-NIREH Bhopal <i>"Multi-Epitope Vaccine design against the structural proteins of Langya Virus."</i> - Leveraged immunoinformatics to design a multi-epitope vaccine targeting 5 structural proteins of the Langya Virus. - Suggested a novel strategy to include epitopes in the construct while maintaining solubility and biocompatibility and published the article as a co-first author.

- Nov – Dec 2023 WINTER INTERNSHIP** | Dr. Abdur Rehman, Biochemistry lab, NISER
"Introduction to Life Cycle of Tetrahymena."
- Learned confocal microscopy and cell imaging using ImageJ.
- May – July 2023 SUMMER SCHOOL** | Dr. S Mirza, IIT Indore
"Principles of Drug Discovery and Development"
- Focus: Understanding the activity of anti-inflammatory drugs and stages of drug discovery.
- May – July 2022 SUMMER INTERNSHIP** | Dr. C S Purohit, Synthetic Chemistry Lab, NISER
"Multi-step Organic synthesis of 2,6-Bis{[bis(2-pyridinylmethyl)amino]methyl}-4-(1,1-dimethylethyl)phenol by aminomethylation."
- Methods: Synthesis, TLC, Column Chromatography, NMR.

CONFERENCES AND CONCLAVE

- **Conferences:**
Quarterly Union for Academic Disclosure (QUAD)
IAS-SRFP 36th Mid-Year meeting 2025
InBix 2025.
- Participated in the **Springer Nature India Research Conclave** (2023)

RELEVANT TECHNICAL SKILLS

- **Computational Structural Biology:** Molecular Docking, MD Simulation (CHARMM/MARTINI), Molecular Modeling, Macromolecular Structure Analysis, Pharmacophore Modeling.
- **Bioinformatics:** Transcriptomics, Proteomics, Network Biology, Multi-Omics
- **Programming:** LINUX, LaTeX, Java, Python, GPU-based Computing, Web tools development.
- **Wet-lab:** qPCR, RT-PCR, Gel electrophoresis, Plasmid/RNA/DNA isolation, Flow-cytometry, Basic X-ray Crystallography, Protein purification, Synthesis, NMR, Column Chromatography.

AWARDS, FELLOWSHIPS AND SCHOLASTIC ACHIEVEMENTS

- **Competitive Exams Qualified:** National Entrance Screening Test 2021 [All India Rank (AIR) 263]; Graduate Aptitude Test in Engineering 2024 [AIR 1173]; Odisha Joint Entrance Examination 2021 [State Rank 3]; Odisha University of Agriculture and Technology 2021 [AIR 81]; National Eligibility cum Entrance Test 2021.
- **Scholarships/Fellowships:** Department of Atomic Energy (DAE)-DISHA Scholarship of 400000 INR(2021-2026); Indian Academy of Sciences-Summer Research Fellowship 30000 INR(May – July 2025).
- **Competitions:** Finalist of BENSPARK Ideathon, IISER Pune [2025]; Excelling ranks in Olympiads (English and Science) [2017-2019]; Won multiple inter-school and intra-college quizzes.

LEADERSHIP AND RESPONSIBILITY

- **Gym Treasurer (2024-2025):** Handled expenditure and bills amounting to almost 50000 INR for NISER Gym Club; Conducted Power-lifting events and served as a coordinator for multiple events.
- **Student Representative:** Member of the Disciplinary Action Committee (DAC) for Students (2025-2026). Contributed actively to policy making and adjudicated cases like substance abuse and harassment to ensure a safe campus environment.

HOBBIES AND EXTRA-CURRICULAR

- **Sports:** Represented NISER in Basketball (IISM 2023) and won inter-batch Basketball tournament (2024). Represented batch in Football (2024) and Badminton (2025) while winning bronze medals.
- **Cultural:** Won multiple dance and slogan writing competitions.
- **Other Interests:** Gym; Reading about novel therapeutic strategies in free time.

REFEREES

Dr. Sindhuprava Rana
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